# Lab 3 Assignment

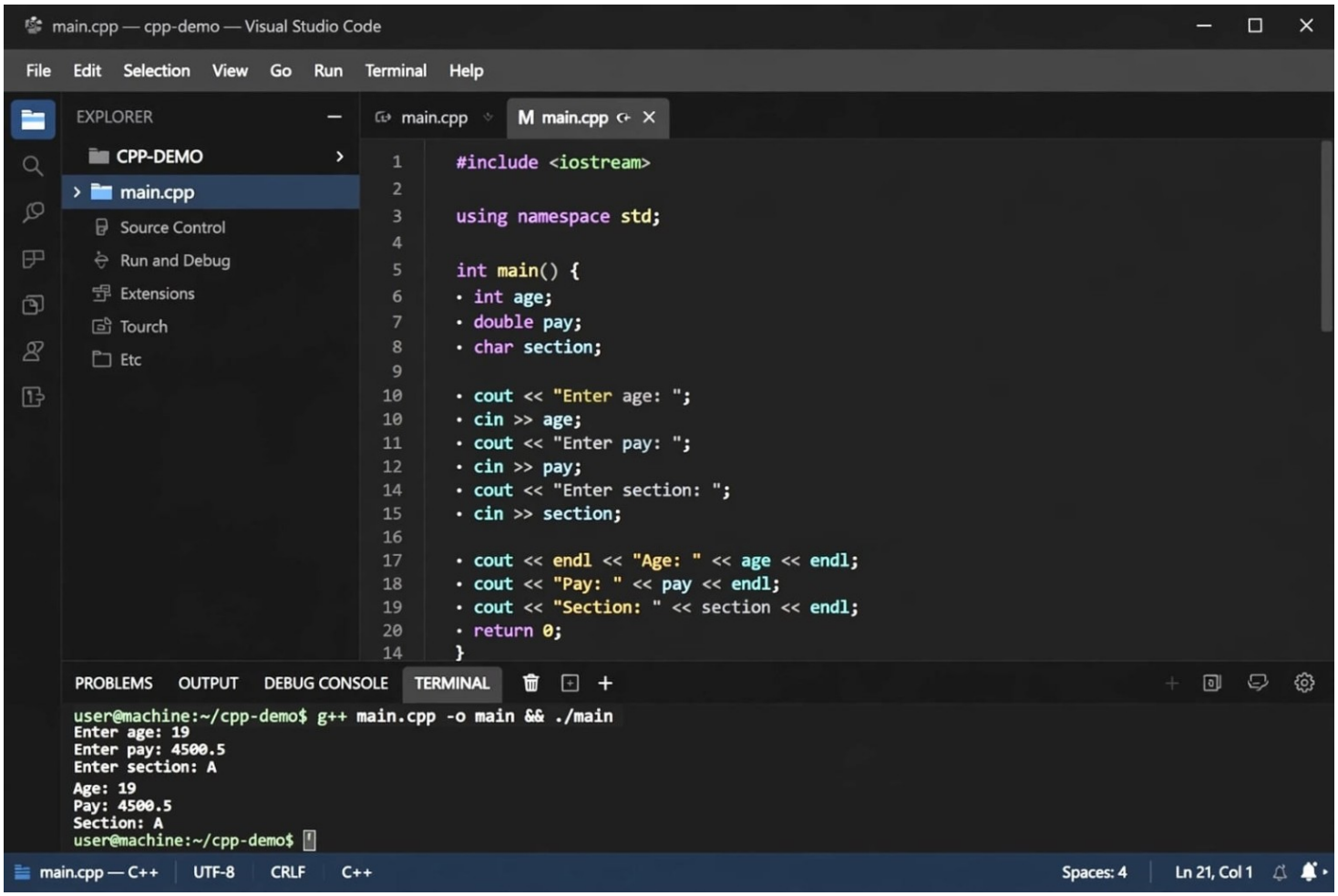
Task 1:-

```
#include <iostream>
using namespace std;

int main()
{
int age;
double pay;
char section;
cout << "Enter age:";
cin >> age;
cout << "Enter section:";
cin >> section;
return 0;
}
```

Output:-

Enter age : 19
Enter pay : 4500.5
Enter section : A

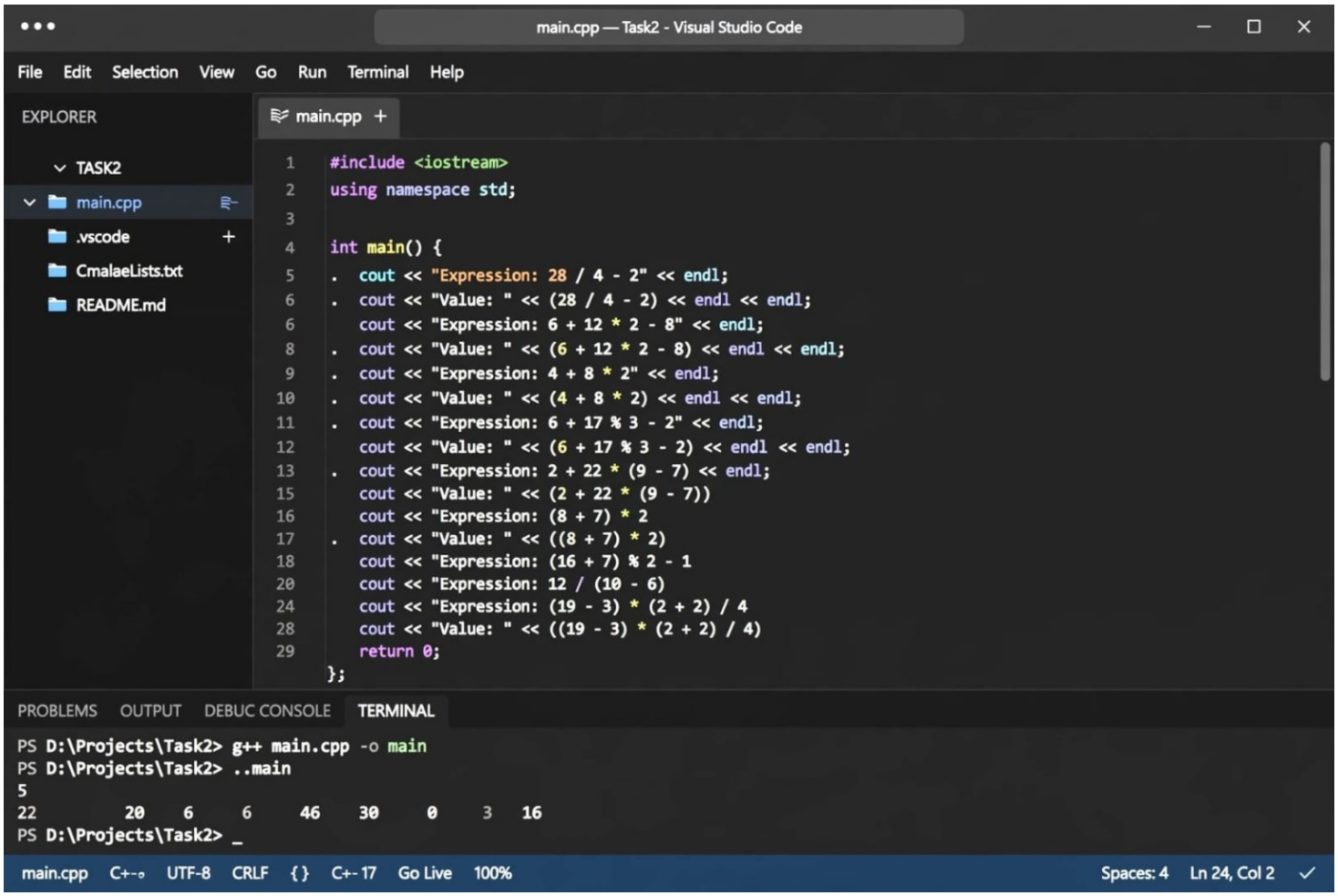

main.cpp — cpp-demo — Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
CPP-DEMO
main.cpp
Source Control
Run and Debug
Extensions
Tourch
Etc
main.cpp
M main.cpp
#include <iostream>
using namespace std;
int main() {
int age;
double pay;
char section;
cout << "Enter age: ";
cin >> age;
cout << "Enter pay: ";
cin >> pay;
cout << "Enter section: ";
cin >> section;
cout << endl << "Age: " << age << endl;
cout << "Pay: " << pay << endl;
cout << "Section: " << section << endl;
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
user@machine:~/cpp-demo$ g++ main.cpp -o main && ./main
Enter age: 19
Enter pay: 4500.5
Enter section: A
Age: 19
Pay: 4500.5
Section: A
user@machine:~/cpp-demo$
main.cpp — C++ UTF-8 CRLF C++
Spaces: 4 Ln 21, Col 1

Task 2:-

```
#include <iostream>
using namespace std;
int main()
{
cout << "Expression: 28 / 4 - 2" << endl;
cout << "Value: " << (28 / 4 - 2) << endl << endl;
cout << "Expression: 6 + 12 * 2 - 8" << endl;
cout << "Value: " << (6 + 12 * 2 - 8) << endl << endl;
cout << "Expression: 4 + 8 * 2" << endl;
cout << "Value: " << (4 + 8 * 2) << endl << endl;
cout << "Expression: 6 + 17 % 3 - 2" << endl;
cout << "Value: " << (6 + 17 % 3 - 2) << endl << endl;
cout << "Expression: 2 + 22 * (9 - 7)" << endl;
cout << "Value: " << (2 + 22 * (9 - 7)) << endl << endl;
cout << "Expression: (8 + 7) * 2" << endl;
cout << "Value: " << (18 + 7) * 2) << endl << endl;
cout << "Expression: (16 + 7) % 2 - 1" << endl;
cout << "Value: " << ((16 + 7) % 2 - 1) << endl << endl;
cout << "Expression: 12 / (10 - 6)" << endl;
cout << "Value: " << (12 / (10 - 6)) << endl << endl;
cout << "Expression: (19 - 3) * (2 + 2) / 4" << endl;
cout << "Value: " << ((19 - 3) * (2 + 2) / 4) << endl;
return 0;
}
```

Output.

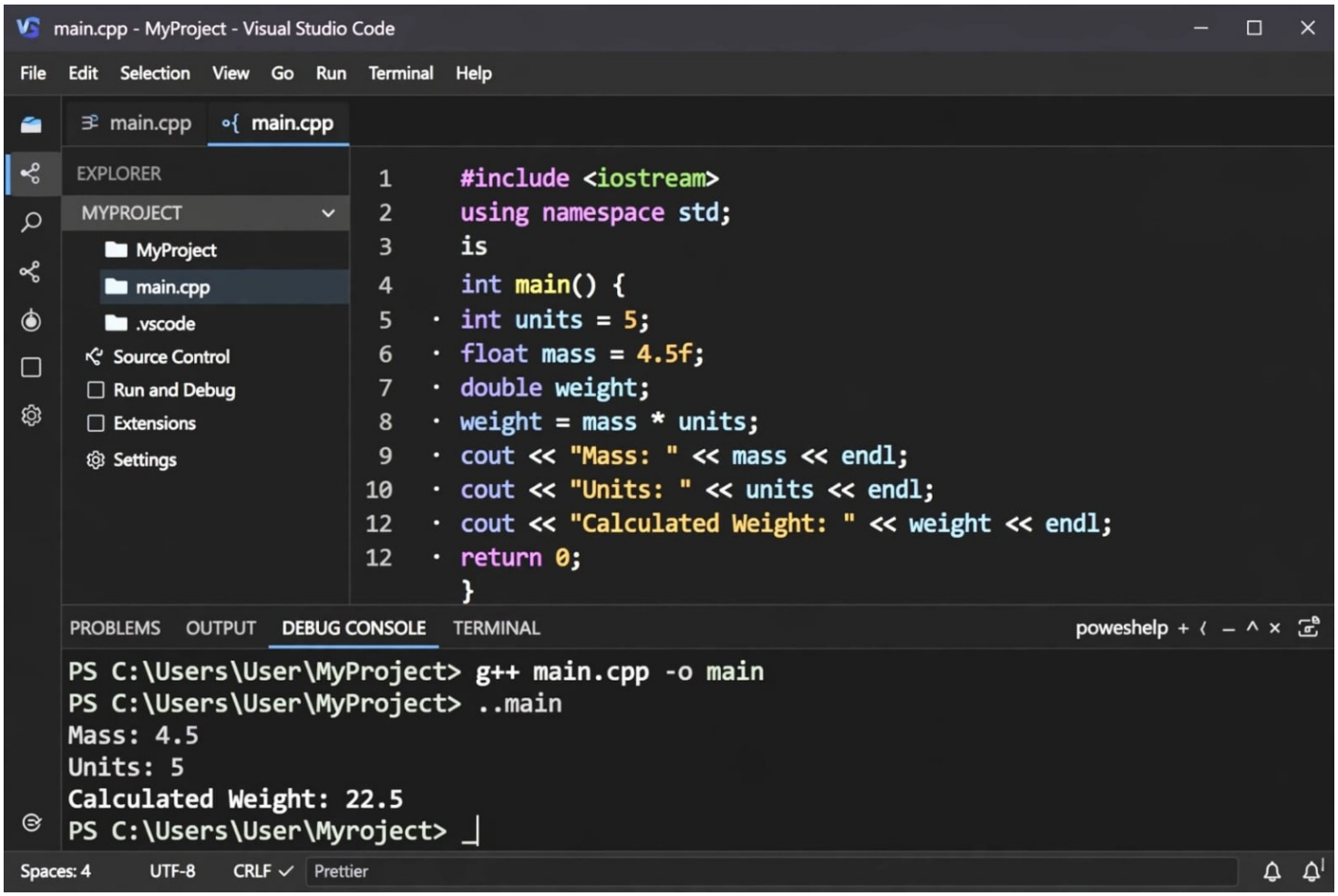
| Expression: | Value: |
|---|---|
| 28/4 - 2 | 5 |
| 6 + 12 * 2 - 8 | 22 |


main.cpp — Task2 - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
TASK2
main.cpp
.vscode
CmalaeLists.txt
README.md
main.cpp +
#include <iostream>
using namespace std;
int main() {
cout << "Expression: 28 / 4 - 2" << endl;
cout << "Value: " << (28 / 4 - 2) << endl << endl;
cout << "Expression: 6 + 12 * 2 - 8" << endl;
cout << "Value: " << (6 + 12 * 2 - 8) << endl << endl;
cout << "Expression: 4 + 8 * 2" << endl;
cout << "Value: " << (4 + 8 * 2) << endl << endl;
cout << "Expression: 6 + 17 % 3 - 2" << endl;
cout << "Value: " << (6 + 17 % 3 - 2) << endl << endl;
cout << "Expression: 2 + 22 * (9 - 7) << endl;
cout << "Value: " << (2 + 22 * (9 - 7))
cout << "Expression: (8 + 7) * 2
cout << "Value: " << ((8 + 7) * 2)
cout << "Expression: (16 + 7) % 2 - 1
cout << "Expression: 12 / (10 - 6)
cout << "Expression: (19 - 3) * (2 + 2) / 4
cout << "Value: " << ((19 - 3) * (2 + 2) / 4)
return 0;
};
PROBLEMS OUTPUT DEBUC CONSOLE TERMINAL
PS D:\Projects\Task2> g++ main.cpp -o main
PS D:\Projects\Task2> ..main
5
22 20 6 6 46 30 0 3 16
PS D:\Projects\Task2> _
main.cpp C+-◦ UTF-8 CRLF { } C+- 17 Go Live 100%
Spaces: 4 Ln 24, Col 2

| | |
|---|---|
| 4+8*2 | 20 |
| 6+17%3-2 | 6 |
| 2+22*(9-7) | 46 |
| (8+7)*2 | 30 |
| (16+7)%2-1 | 0 |
| 12/(10-6) | 3 |
| (19-3)*(2+2)/4 | 16 |

Task 3:-

```
#include <iostream>
using namespace std;
int main()
{
int units = 5;
float mass = 4.5f;
double weight;
weight = mass * units;
cout << "Mass: " << mass << endl;
cout << "Units: " << units << endl;
cout << "Calculated Weight: " << weight << endl;
return 0;
}
```


main.cpp - MyProject - Visual Studio Code
File Edit Selection View Go Run Terminal Help
main.cpp
main.cpp
EXPLORER
MYPROJECT
MyProject
main.cpp
.vscode
Source Control
Run and Debug
Extensions
Settings
#include <iostream>
using namespace std;
is
int main() {
int units = 5;
float mass = 4.5f;
double weight;
weight = mass * units;
cout << "Mass: " << mass << endl;
cout << "Units: " << units << endl;
cout << "Calculated Weight: " << weight << endl;
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
poweshelp
PS C:\Users\User\MyProject> g++ main.cpp -o main
PS C:\Users\User\MyProject> ..main
Mass: 4.5
Units: 5
Calculated Weight: 22.5
PS C:\Users\User\Myroject>
Spaces: 4 UTF-8 CRLF Prettier

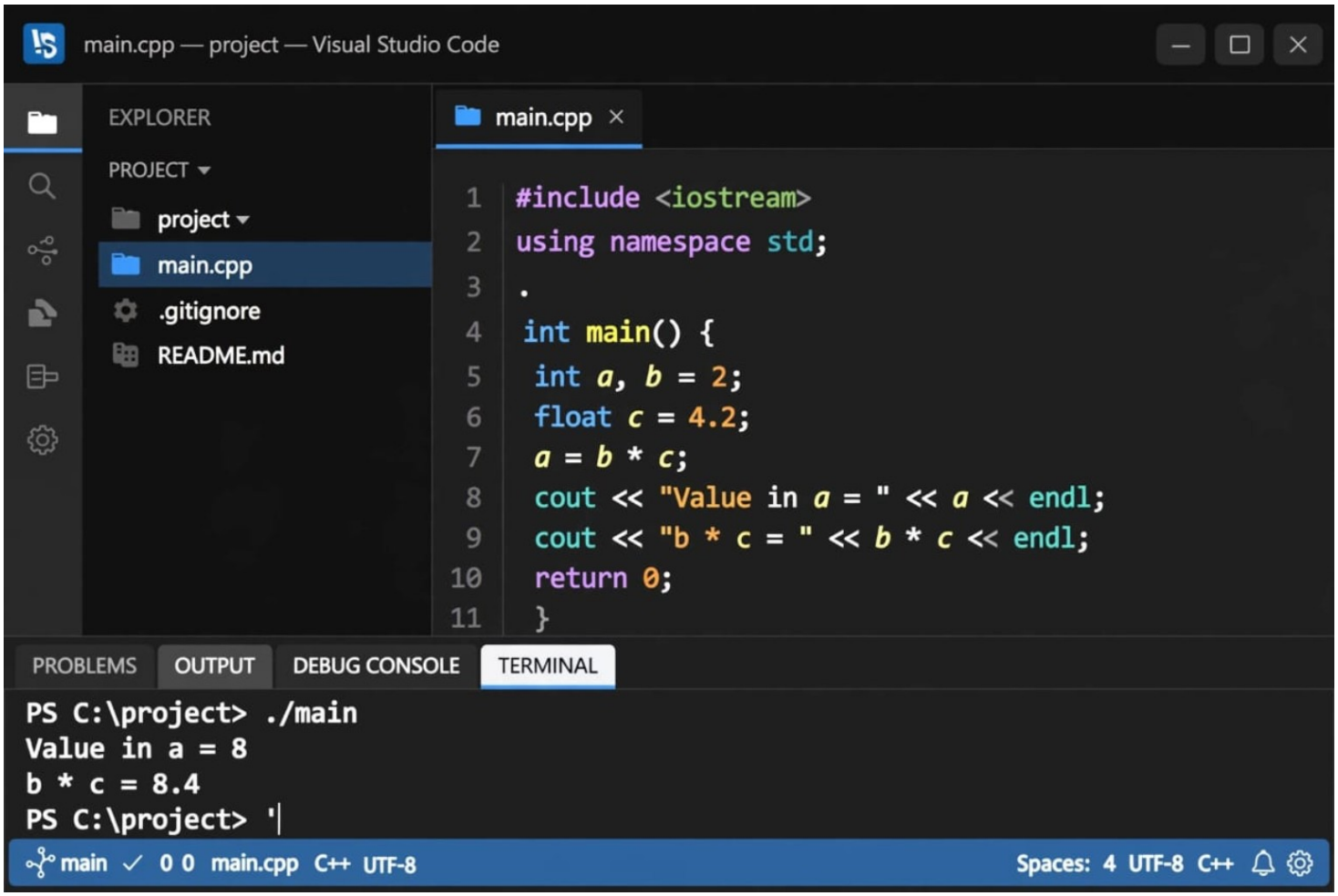
Task 4:-

```
#include <iostream>
using namespace std;
int main()
{
int a, b = 2;
float c = 4.2;
a = b * c;
cout << "Value in a = " << a << endl;
cout << "b * c = " << b*c << end;
return 0;
}
```

Output:-

Value in a = 8

b * c = 8.4


main.cpp — project — Visual Studio Code
EXPLORER
PROJECT
project
main.cpp
.gitignore
README.md
main.cpp
#include <iostream>
using namespace std;
.
int main() {
int a, b = 2;
float c = 4.2;
a = b * c;
cout << "Value in a = " << a << endl;
cout << "b * c = " << b * c << endl;
return 0;
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PS C:\project> ./main
Value in a = 8
b * c = 8.4
PS C:\project> '
main
0 0
main.cpp
C++
UTF-8
Spaces: 4
UTF-8
C++

Task 5:-

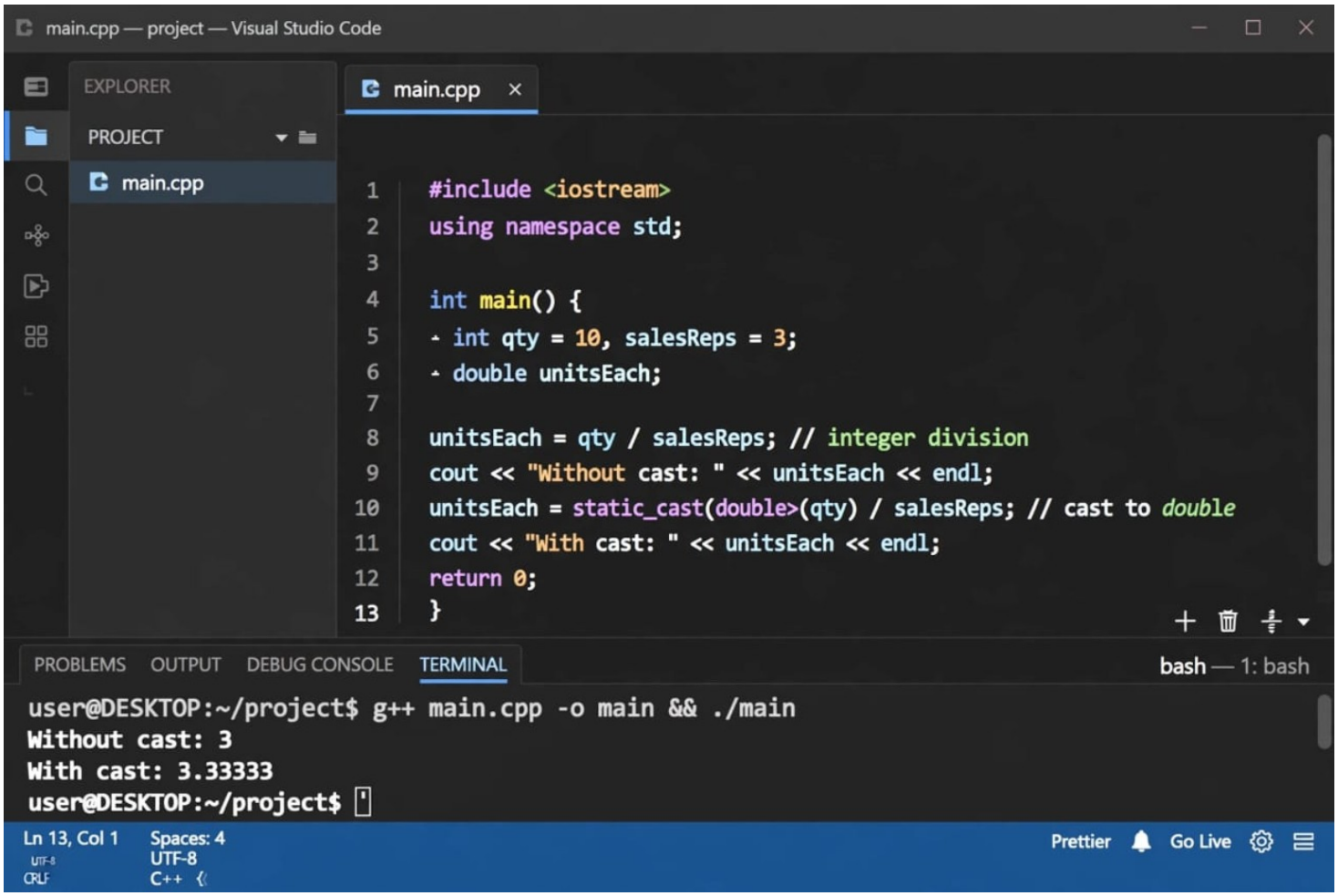
```
#include <iostream>
using namespace std;
int main()
{
 int qty = 10, salesReps = 3;
 double unitsEach;
 unitsEach = qty / salesReps; // integer division
 cout << "Without cast : " << unitsEach << endl;
 unitsEach = static_cast (double > (qty) / salesReps; //
 cout << "With cast: " << unitsEach << endl;
 return 0;
}
```

cast to double

Output :-

Without cast : 3

With cast : 3.33333

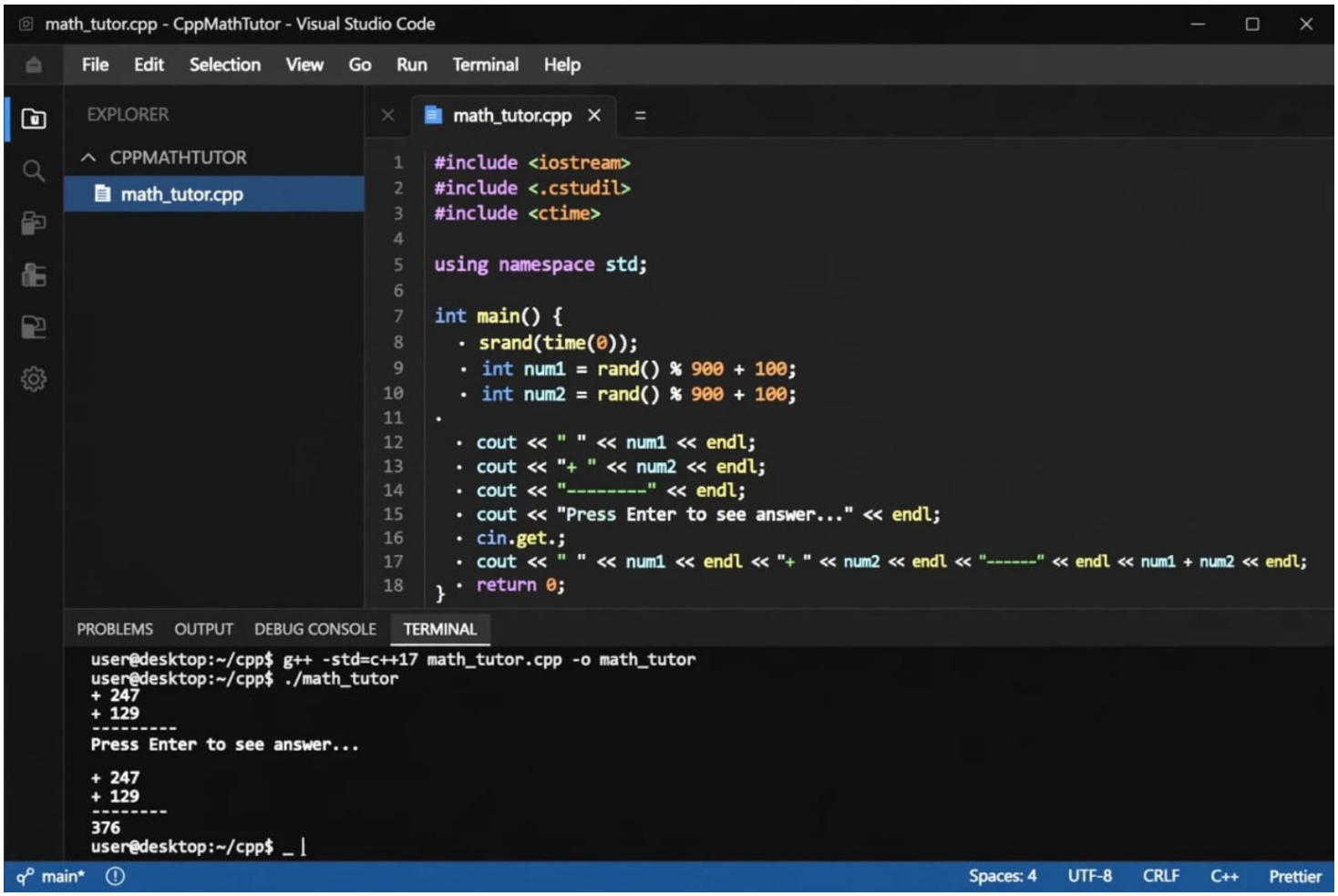

main.cpp — project — Visual Studio Code
EXPLORER
PROJECT
main.cpp
main.cpp
#include <iostream>
using namespace std;
int main() {
int qty = 10, salesReps = 3;
double unitsEach;
unitsEach = qty / salesReps; // integer division
cout << "Without cast: " << unitsEach << endl;
unitsEach = static_cast(double>(qty) / salesReps; // cast to double
cout << "With cast: " << unitsEach << endl;
return 0;
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
bash — 1: bash
user@DESKTOP:~/project$ g++ main.cpp -o main && ./main
Without cast: 3
With cast: 3.33333
user@DESKTOP:~/project$
Ln 13, Col 1
Spaces: 4
UTF-8
CRLF
C++
Prettier
Go Live

Task 6:-

```
#include<iostrem>
#include<.cstudil>
#include<ctime>
using namespace std;
int main()
{
srand(time(0));
int num1=rand() % 900+100;
int num2=rand() % 900+100;
cout<<" "<<num1<<endl;
cout<<"+ "<<num2<<endl;
cout<<"-----"<<endl;
cout<<"Press Enter to see answer---"<<endl;
cin get.;
Cout<<" "<<num1<<"+"<<num2<<endl<<"----"endl
<<num1+ nm2<<endl;
return 0;
}
```

Output

```
+ 247
+ 129
-------
  376
```


math_tutor.cpp - CppMathTutor - Visual Studio Code
File Edit Selection View Go Run Terminal Help
EXPLORER
CPPMATHTUTOR
math_tutor.cpp
math_tutor.cpp
#include <iostream>
#include <.cstudil>
#include <ctime>
using namespace std;
int main() {
srand(time(0));
int num1 = rand() % 900 + 100;
int num2 = rand() % 900 + 100;
cout << " " << num1 << endl;
cout << "+ " << num2 << endl;
cout << "---------" << endl;
cout << "Press Enter to see answer..." << endl;
cin.get.;
cout << " " << num1 << endl << "+ " << num2 << endl << "------" << endl << num1 + num2 << endl;
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
user@desktop:~/cpp$ g++ -std=c++17 math_tutor.cpp -o math_tutor
user@desktop:~/cpp$ ./math_tutor
+ 247
+ 129
---------
Press Enter to see answer...
+ 247
+ 129
--------
376
user@desktop:~/cpp$ _
main*
Spaces: 4 UTF-8 CRLF C++ Prettier